Semester production README & architecture guide

Developer, deployment and activation review pack

Inspected source baseline: 4b4f8ca742242132f5b2dd254d23422ae70eaa7a

Prepared October 11, 2026. Source definitions and proposed requirements are distinct. No live API probes, full app tests, migration or deployment were performed.

The PDF is a static report. The accompanying HTML API catalog supports filtering and column sorting; CSV supports spreadsheet analysis.

Executive launch decision

Entire-platform activation is NOT established. At the inspected source baseline, approved institutional adapters are absent. Registration-readiness routes exist but their handler is not wired in the inspected startup. Health readiness does not establish business capability readiness.

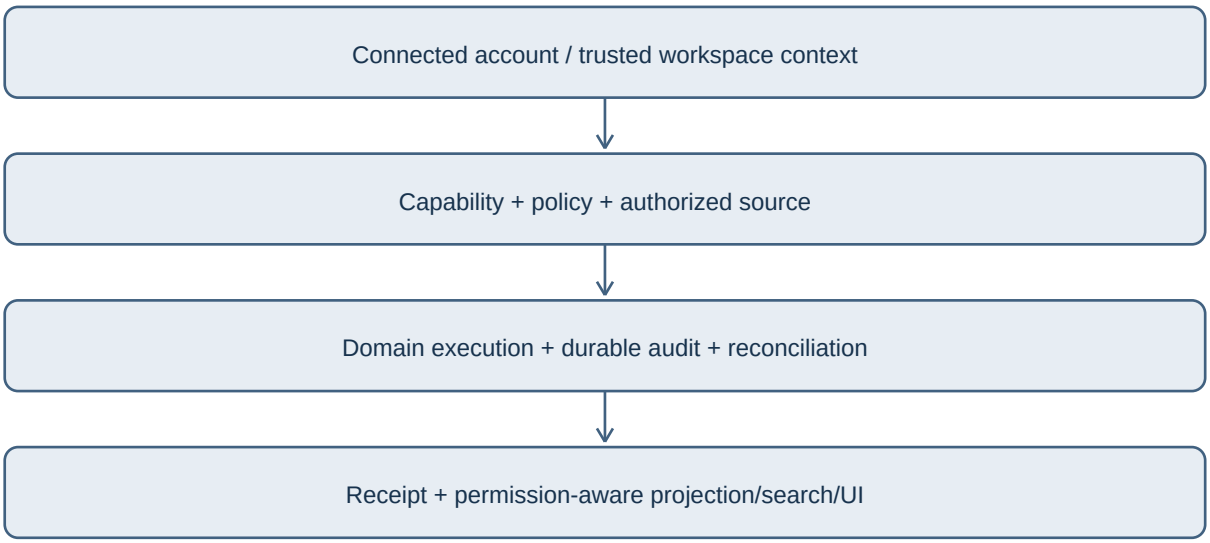
This delivery supplies a launch specification, exact additive patch, API inventory and reviewed orchestration. It does not implement missing domains or grant institutional authority.

1. Evidence and architecture boundaries

Inspected: gateway.ts, start.ts, auth.ts and adapters.ts under app/server/institution. Prior review: SETUP.md, ROLLBACK.md, CI and student/browser harnesses. Some gateway/harness reads were truncated; this is not a full code audit.

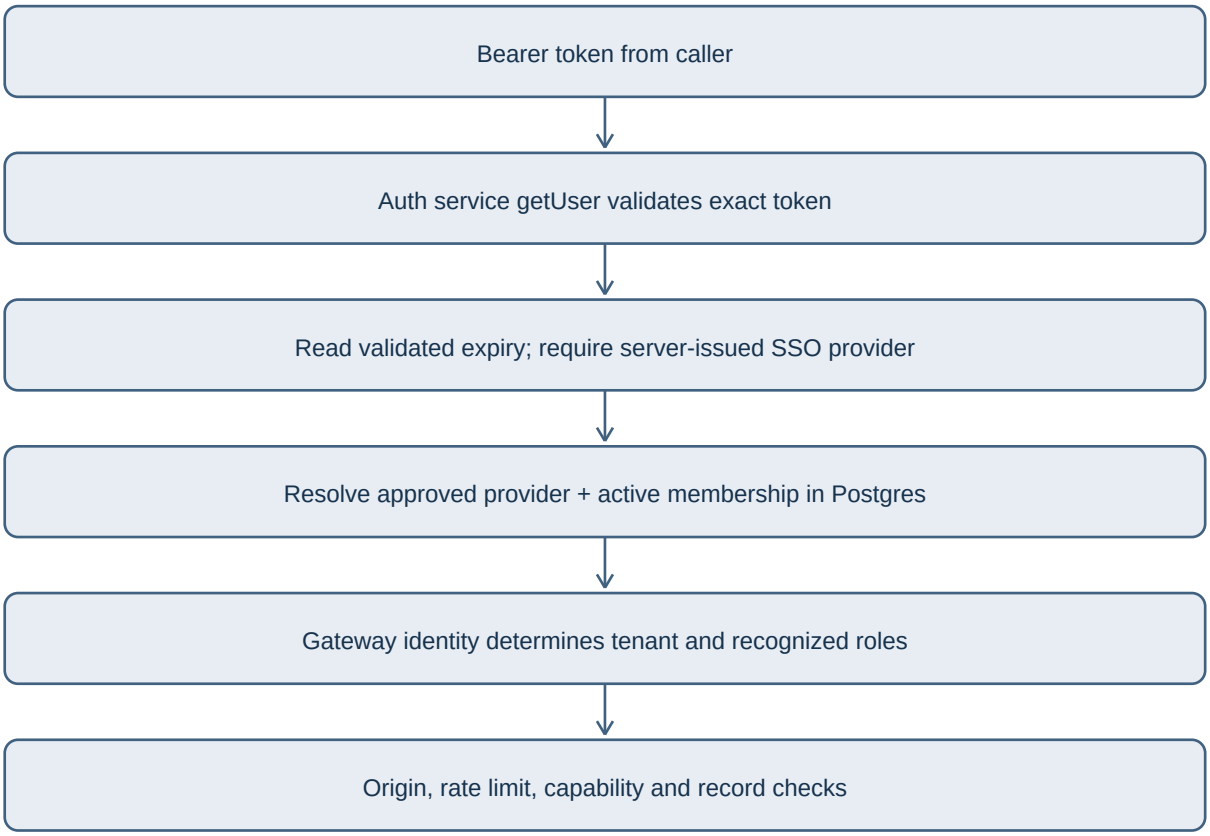
Design catalogs describe the intended domain and screen scope, including authority transitions. Their routes are not automatically HTTP APIs and their status statements are not current production verification.

One account and shared shell connect personal, institution and company contexts. They do not share blanket access. Company reporting uses approved business/aggregate records, not unrestricted education records.



2. Institutional authentication flow

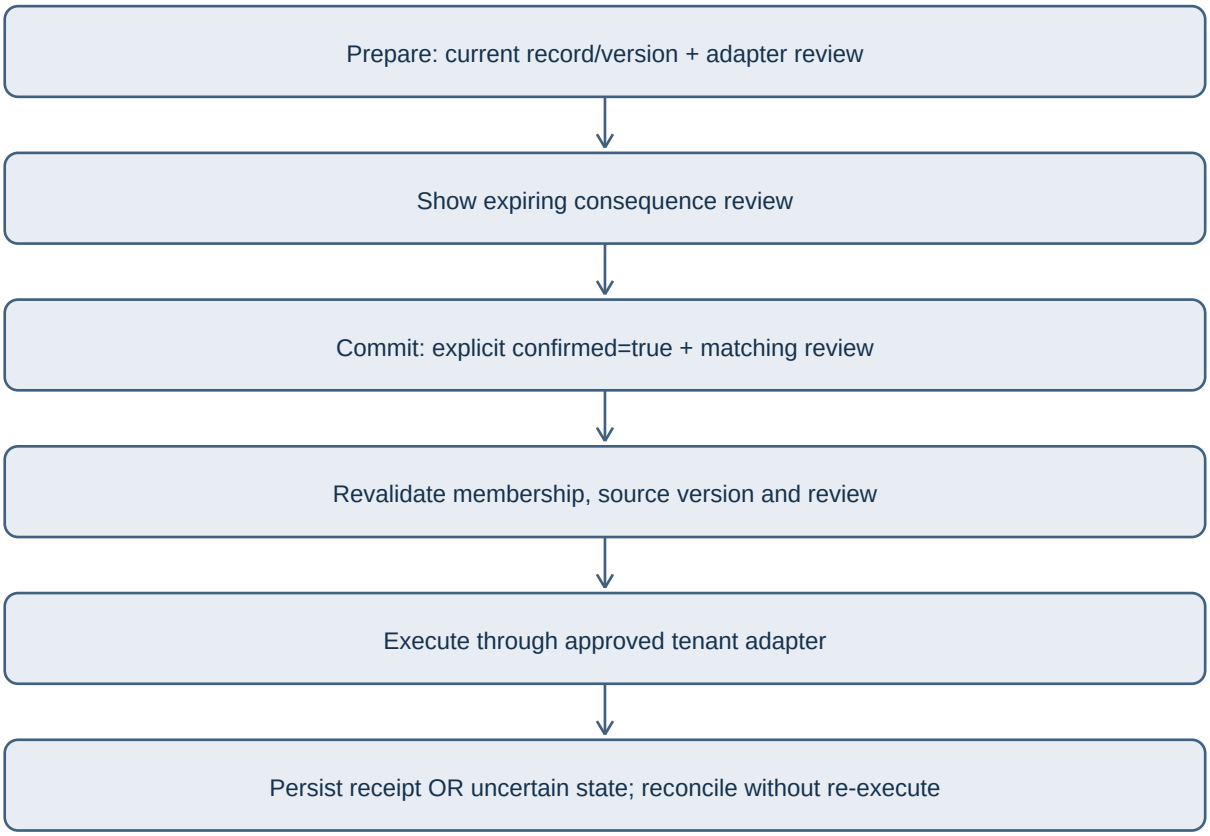
Source: app/server/institution/auth.ts and start.ts. Institutional identity is distinct from ordinary personal account sign-in. Production startup uses a membership resolver; the legacy metadata helper is for fixtures/sandbox.



Missing bearer yields 401; rejected institutional identity yields 403 in the inspected gateway. Revocation/session behavior must be verified against the live auth configuration; source comments are not measured guarantees.

3. Consequential action flow

Source routes: POST /actions/prepare, /actions/commit, /actions/reconcile. Prepare performs validation/review and journal persistence, not the institutional execute.



The full commit tail was not included in the truncated gateway return. Detailed concurrency/idempotency claims require inspecting the journal/adapter/tests. Reconciliation remains allowed in read-only mode.

4. Source-defined API catalog (1-5)

Paths are relative to the gateway base; no deployed gateway origin is verified here. This is 14 observed routes, not the entire platform API census.

GET /health/live

Process liveness

Auth: Public; origin rules | Gate: Not business readiness

GET /health/ready

Journal/custom readiness; adapter count; AI/read-only state

Auth: Public; origin rules | Gate: 200 does not establish active adapters

GET /health

Readiness alias

Auth: Public; origin rules | Gate: Same limitation as /health/ready

GET /v1/auth/config

SSO presentation configuration

Auth: Public; origin rules | Gate: Returns disabled when unavailable

GET /status

Connection status, roles and institution

Auth: Bearer + institutional membership | Gate: Installed tenant/area adapters

4. Source-defined API catalog (6-10)

Paths are relative to the gateway base; no deployed gateway origin is verified here. This is 14 observed routes, not the entire platform API census.

GET /records

Area-scoped records; area/search/cursor query

Auth: Bearer + institutional membership | Gate: Approved adapter and canRead

POST /actions/prepare

Validate record/version; prepare expiring review

Auth: Bearer + institutional membership | Gate: Approved adapter and canWrite

POST /actions/commit

Confirm review; recheck current access/record

Auth: Bearer + institutional membership | Gate: Explicit true, valid review, compatible adapter

POST /actions/reconcile

Resolve uncertain outcome; not repeat execution

Auth: Bearer + institutional membership | Gate: Adapter reconciliation; allowed in read-only mode

GET /v1/intelligence/policy

Read institutional AI policy

Auth: Bearer + institutional membership | Gate: Configured intelligence service

4. Source-defined API catalog (11-14)

Paths are relative to the gateway base; no deployed gateway origin is verified here. This is 14 observed routes, not the entire platform API census.

POST /v1/intelligence/respond

Policy-controlled intelligence request

Auth: Bearer + institutional membership | Gate: Server provider, policy, sources and budget

POST /v1/intelligence/actions/:id/confirm

Confirm bounded intelligence action

Auth: Bearer + institutional membership | Gate: Service confirmation rules/current access

POST /v1/registration-readiness/evaluations

Start readiness evaluation; 202 receipt

Auth: Bearer + refreshed membership | Gate: registrationReadiness handler; omitted in inspected start.ts

POST /v1/registration-readiness/evaluations/:id/evaluate

Evaluate existing request; empty object body

Auth: Bearer + refreshed membership | Gate: Handler required; result 200 or 202

5. Template-to-endpoint mapping

- Institution console: GET /status + GET /records; adapter and membership gates. Source route exists, real records not established.

- Registrar readiness: POST /v1/registration-readiness/evaluations and /:id/evaluate; gateway handlers exist, inspected start.ts does not pass registrationReadiness dependency. Wiring, service schema and institution approval remain required.

- Consequential action/registrar: POST /actions/prepare, /commit, /reconcile. No generic create-course API is inferred from these routes.

- Course Studio: domain target needs authoring, immutable release, section scopes, submissions, feedback and grades. Gateway adapter boundary is a potential integration surface, not verified Course Studio CRUD endpoints. No new route name invented here.

- AI answer/coplan: GET /v1/intelligence/policy, POST /respond, POST /actions/:id/confirm. Provider/policy/data/budget gates remain.

- Device/privacy: app preference, OS permission, consent and session revocation are different contracts. Gateway auth revalidation is only one layer; privacy CRUD endpoints not verified in this inspection.

- Board/investor: company-only sources and immutable approved aggregates; financial reporting/data-room endpoints not verified here. Do not send raw student records to company reports.

- Company deck/policy/training/research: controlled documents, not necessarily APIs. Stable IDs/version/review/disclosure and ownership required.

Exact new code changes in this delivery: documentation, API catalog and reviewed test orchestration. No runtime gateway, adapter, auth, DDL or feature enablement modifications. A production implementation diff requires inspecting the relevant existing service, schema and test seams first.

6. Exact diff review and orchestrator findings

Status: revise before claiming launch evidence. Review based on exact source shown in the prior conversation; prior generated archive was not available in this execution environment.

1. Original diff adds eight documentation/orchestration files only. No product, API or database capability is implemented by that diff.

2. Previous git-apply check was against a clean temporary repository, not the actual Semester worktree. Baseline 4b4f8ca742242132f5b2dd254d23422ae70eaa7a must be rechecked/rebased before application.

3. Original orchestrator handled timeout but not missing npm/OS launch errors. Revised runner records unavailable/launch_error.

4. Original setup failures printed text without a report. Revised CLI persists a failed report where possible; persistence failure exits 2.

5. Original runner passed the local service key to all child suites. Revised runner removes that key except for account-sync. Other inherited credentials are not sandboxed; use a clean ephemeral test environment.

6. Child stdout/stderr are now discarded by default, preventing unbounded in-memory captured logs and accidental report credential persistence. Individual child scripts must be run directly in an approved environment for diagnostics.

7. Loopback URL does not prove all backend/provider requests are local. Keep the existing account-sync guard and inspect actual build configuration.

8. Exit-zero trusts child exit semantics. This runner cannot independently verify every browser step; preserve underlying suite evidence. It never labels a selected student scope as full-platform readiness.

9. Full Course Studio, institutional authority, company reporting and production rollback are excluded. They need mapped tests and fixtures.

10. Missing test prerequisites must block, never skip or fabricate success. New unit tests validate orchestration only.

This revised patch is a new additive review package under docs/launch-control-guide. It does not automatically replace the prior proposed files, change package scripts or integrate CI. Choose one canonical runner during repository review; do not deploy parallel competing launch mechanisms.

7. Production migration walkthrough

1. Confirm sole project, production mapping and executor. Prior inventories contain multiple projects; names alone are insufficient.

2. Compare repository migration hashes, live ledger, and object/policy fingerprints. Stop on unexplained drift.

3. Design an additive, backward-compatible schema/API change. No DDL is included in this patch because exact runtime schema need is not yet verified.

4. Rehearse with disposable PostgreSQL and synthetic preexisting state. Test old/new app compatibility, role/tenant denials, lock impact, retries and recovery.

5. Approve exact SQL/target/hash, accountable owner and restore evidence before execution or any merge that triggers migration.

6. Let one verified executor apply; inspect ledger and actual objects, then functions/app and observed domain journeys.

7. Activate approved domains. Destructive cleanup is a later reviewed release. App rollback does not roll back schema or lost data.

8. Developer setup and environment

Use the existing root lockfile and current environment guide. Node >=22 is the prior inspected root requirement.

Commands: npm ci; npm run dev --workspace app; npm run test --workspace app; npm run build --workspace app; npm run registry:validate; npm run platform-control:test; npm run release:check; npm run release:report. Not executed here.

Gateway startup binds 127.0.0.1, default port 8787. Exposure/TLS topology must be explicit. SEMESTER_APP_ORIGIN must be exact; SEMESTER_JOURNAL_KEY is server-only 32-byte hex.

Auth URL/public key/service key, membership loader, SSO configuration, journal/store and AI provider/budget values need approved server configuration. Never put secrets into VITE variables.

Source-defined startup offers an explicitly labelled sandbox; it is not an approved institution adapter. Do not use sandbox responses as production evidence.

9. Onboarding and workspace acceptance

Student: explain local versus synced work; create/confirm account or continue locally; import syllabus under approved processing; review sources/dates; save; create a useful action; verify persistence; optional sharing/device permissions separately.

Institution: sponsor/data steward; approved tenant/database/identity mapping; agreements; explicit grants; sources/freshness; synthetic test identities; reconciliation; role training; approval.

Faculty: section/teaching delegation; author and preview; accessibility/AI rule review; controlled publish; submission receipt/review/release; official grade authority only after approval.

Staff: assigned scope; source/authority inspection; consequence preview; approved command; receipt/escalation. Investor: invite, identity checks, audience scope, read-only publications, expiry.

Completion requires authorized home/action, denials, truthful source and sync, reversible preferences and reachable support. Offboarding must honor access revocation and official retention.

10. Deployment and rollback execution

PR tests + registry validity -> isolated preview + schema rehearsal -> immutable release manifest -> blocking release check + approval -> database expand -> functions/API -> app -> observed role journeys -> domain activation.

Existing Pages, Vercel, Edge Functions and Supabase workflows need one verified mapping/executor. No new competing deployment automation is introduced by this patch.

Rollback: authorized containment -> tested compatible prior app/function -> schema check -> restore selected delivery pointer/ref -> actual page/cache/service-worker verification -> auth/read/write/reconciliation -> incident record.

Do not auto-run destructive down migrations. Server read-only is distinct from UI read-only; inspected gateway intentionally permits reconciliation. Automatic rollback needs tested target and preauthorization; otherwise alert and human control.

No newly measured recovery targets or incident thresholds are asserted. Approve thresholds from actual baseline evidence.

11. Test execution and remaining coverage

Revised runner composes smoke:cold, smoke:a11y, smoke:golden and optionally smoke:sync. Only loopback URL inputs. Account tests require separately configured local auth build and ephemeral service key. Runner does not provision prerequisites.

15 unit tests were executed for the revised orchestrator. They mock child suites, so they prove orchestration behavior only. Application browser/account tests remain unexecuted in this delivery.

Remaining production coverage: faculty submission/release, section delegation, registration stale/unknown outcome, consent/search/AI revocation, board/data-room disclosure, tenant isolation, durable workers, migration compatibility and restore.

Selected scope pass is never global readiness. Underlying test outputs and release/environment fingerprints remain mandatory evidence.

12. Full ecosystem activation register

For every domain: owner, capability IDs, implementation refs, role scopes, data authority, source/freshness, UI states, policy, test evidence, deployments/config hashes, institution approval, runbook, reconciliation, rollback and activation receipt.

Required families: individual planning/workspace; learning/assessment; registration/records; advising/support; campus/community; student finance; career/alumni; family; institutional governance; integration/migration; trust; company operations; developer platform; data/graph; regional expansion.

Preserve comprehensive scope. Do not label prototype screens live or narrow the catalog to hide unfinished workflows. Finish/harden existing connected workflows before adding duplicates.

Launch owners: engineering release owner; database owner; identity/security owner; domain owners; privacy/accessibility reviewers; finance/disclosure owner; institutional sponsor; operations incident owner. Actual people must be assigned.

Source basis: project native domain/screen catalogs plus inspected repository files. Design authority gates are requirements, not passing results.